

Mandatory Assignment 1

Sander Lydom & Sigurd Sørensen

03 August, 2026

1 Market Data and Excess Returns

We download monthly S&P 500 returns and 13-week T-Bill yields from Yahoo! Finance for January 2000 to February 2026. The T-Bill annualized yield is converted to a monthly risk-free rate via $r_{f,t} = (1 + \text{IRX}_t/100)^{1/12} - 1$, and monthly excess market returns are computed as $\tilde{r}_{m,t} = r_{m,t} - r_{f,t}$.

Monthly mean excess return:	0.0045		Annualized: 0.0545
Monthly variance:	0.001903		Annualized: 0.022841

The estimated annualized mean excess return is `np.float64(0.0545)` and the annualized variance is `np.float64(0.022841)`, based on 313 monthly observations.

2 Synthetic Asset Universe

We construct $N = 50$ synthetic assets from the single-factor model:

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_{\varepsilon,i}^2),$$

where $\beta_i \stackrel{iid}{\sim} \text{Uniform}(0.5, 1.5)$ and $\sigma_{\varepsilon,i} \stackrel{iid}{\sim} \text{Uniform}\left(\frac{0.10}{\sqrt{12}}, \frac{0.30}{\sqrt{12}}\right)$. The true parameters are:

$$\mu_i = \beta_i \mu_m, \quad \Sigma = \sigma_m^2 \beta \beta' + \text{diag}(\sigma_{\varepsilon}^2).$$

These are treated as the true data-generating process throughout.

3 Efficient Frontier Function

The mean-variance efficient frontier is computed analytically using the closed-form solution. For a given target return $\bar{\mu}$, the frontier portfolio weights are:

$$\omega(\bar{\mu}) = \frac{C - B\bar{\mu}}{D}\Sigma^{-1}\iota + \frac{A\bar{\mu} - B}{D}\Sigma^{-1}\mu,$$

where $A = \iota'\Sigma^{-1}\iota$, $B = \iota'\Sigma^{-1}\mu$, $C = \mu'\Sigma^{-1}\mu$, and $D = AC - B^2$. The tangency portfolio is proportional to $\Sigma^{-1}\mu$, scaled to sum to one. Linear systems are solved numerically rather than inverting Σ explicitly for numerical stability.

4 True Efficient Frontier

Figure 1 shows the efficient frontier and tangency portfolio computed from the true parameters μ and Σ .

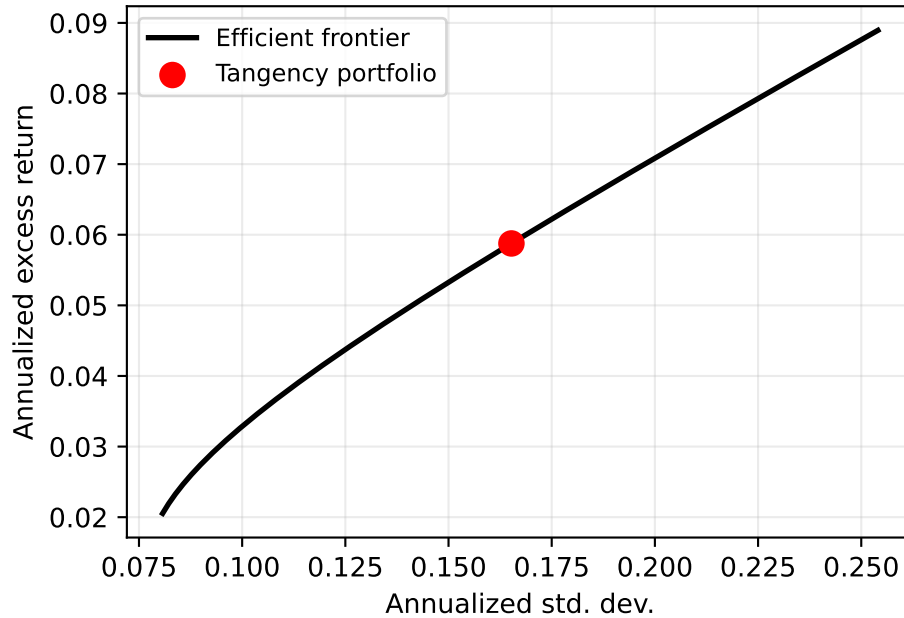


Figure 1: Mean-variance efficient frontier based on true parameters. The red dot marks the tangency portfolio. All values are annualized.

The true tangency portfolio achieves an annualized Sharpe ratio of `np.float64(0.356)`.

5 Estimation Error and the Plug-In Frontier

Using the simulation framework, we draw $S = 100$ samples of size T from the true DGP and compute the plug-in frontier for each sample. Figure 2 shows results for $T \in \{60, 120, 240, 600\}$ months.

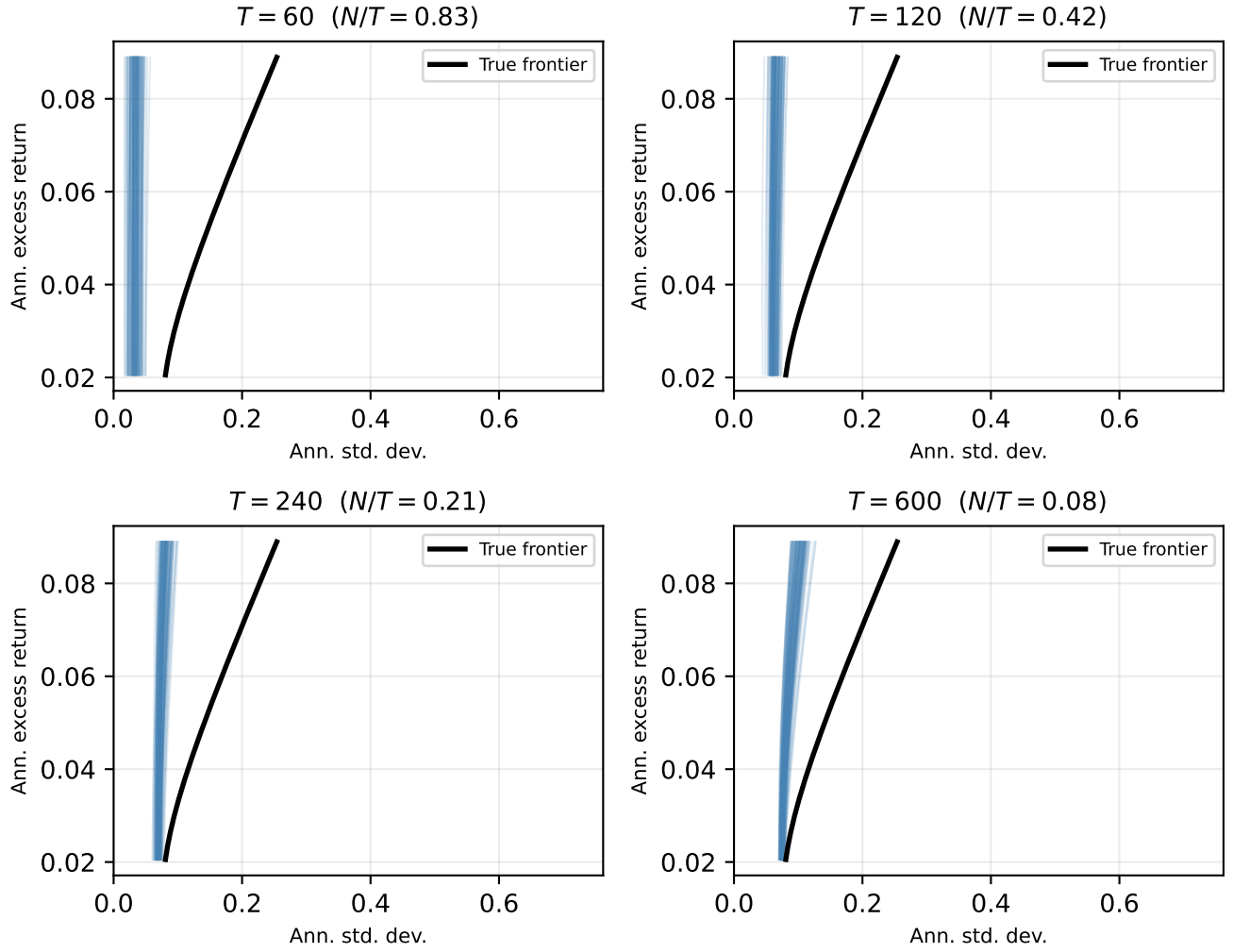


Figure 2: Plug-in efficient frontiers (blue, 100 simulations) versus the true frontier (black) for varying sample sizes T . The ratio N/T governs estimation error.

The key driver of estimation error is the ratio N/T . When T is small relative to N , the sample covariance matrix is poorly conditioned and both $\hat{\mu}$ and $\hat{\Sigma}$ are noisy. Plug-in frontiers systematically lie to the left of the true frontier — overstating attainable Sharpe ratios. As T grows, the plug-in frontiers converge to the true frontier.

6 Out-of-Sample Performance of the Plug-In Tangency Portfolio

For each T , we simulate $S = 500$ samples, estimate plug-in tangency weights $\hat{\omega}_t$, and evaluate their Sharpe ratio using the true parameters. Figure 3 shows the distribution of out-of-sample Sharpe ratios.

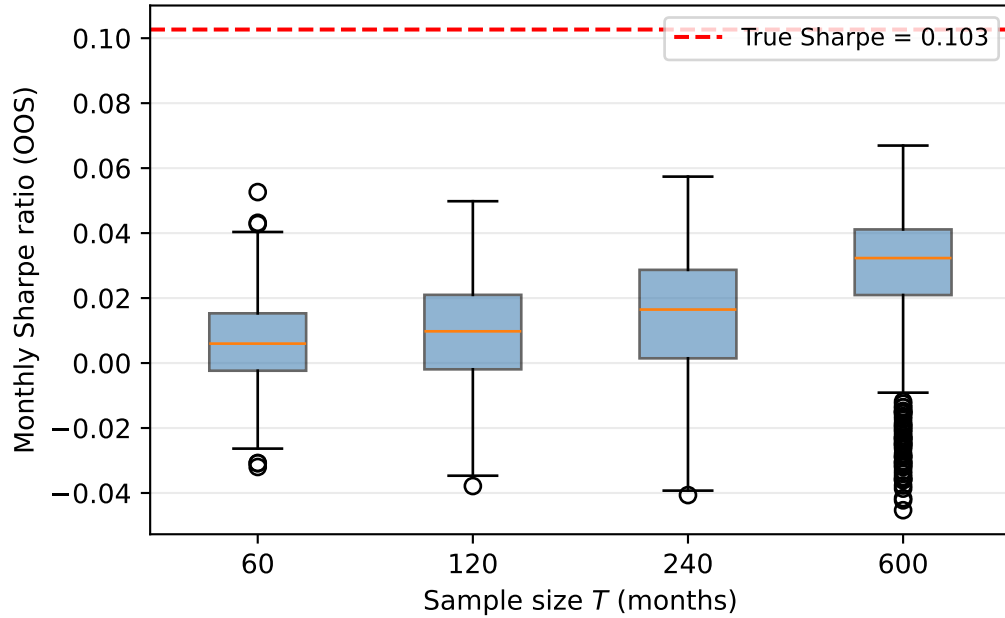


Figure 3: Out-of-sample monthly Sharpe ratio of the plug-in tangency portfolio across 500 simulations. The red dashed line is the true tangency Sharpe ratio.

At $T = 60$ (i.e. $N/T \approx 0.83$), the median out-of-sample Sharpe ratio is far below the true optimum and the distribution is highly dispersed, including negative values. Performance improves substantially as T increases, consistent with ?. The economic loss from estimation error is severe at realistic sample sizes.

7 Alternative Portfolio Strategies

We consider four strategies alongside the plug-in tangency portfolio:

Equal weight ($1/N$). Assigns $\omega_i = 1/N$ to each asset. This ignores all parameter estimates entirely, so estimation error cannot compound. It serves as a robustness benchmark ?.

Global minimum variance (GMV). Solves $\min_{\omega} \omega' \Sigma \omega$ subject to $\omega' \iota = 1$. By avoiding estimation of expected returns — the noisiest input — GMV is substantially more robust than the plug-in tangency portfolio.

Volatility parity. Sets $\omega_i \propto 1/\hat{\sigma}_i$, equalizing each asset’s standalone risk contribution. This requires only diagonal elements of $\hat{\Sigma}$, making it more robust than GMV to off-diagonal estimation error.

Factor portfolio. Sets $\omega_i \propto \beta_i$, allocating in proportion to market exposure. In a single-factor world this is theoretically motivated: beta is the only priced characteristic, and the strategy avoids estimating the full $N \times N$ covariance matrix.

Figure 4 compares all strategies by median out-of-sample Sharpe ratio across sample sizes.

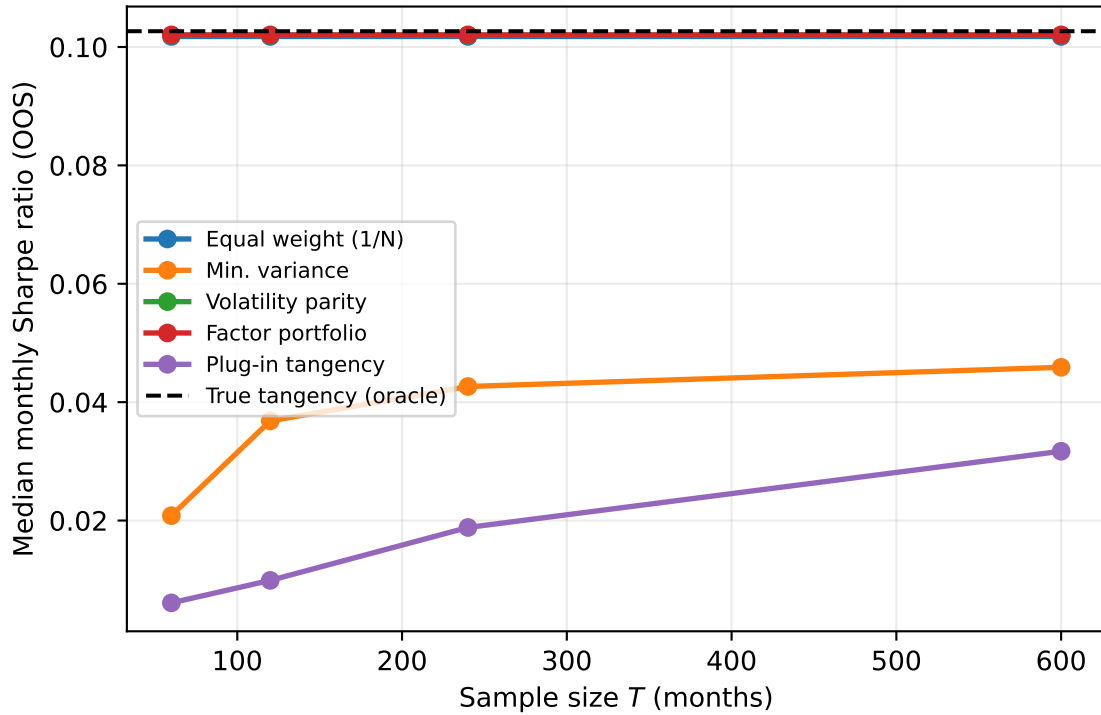


Figure 4: Median out-of-sample monthly Sharpe ratio across 500 simulations for each strategy. The dashed line is the true tangency (oracle) Sharpe ratio.

At small T , the plug-in tangency portfolio is the worst performer — its unrestricted reliance on noisy $\hat{\mu}$ and $\hat{\Sigma}$ causes extreme weight concentration. The factor portfolio performs best at all sample sizes in this simulation, exploiting the known single-factor structure. Strategies that avoid estimating expected returns (GMV, volatility parity, equal weight) consistently outperform the plug-in tangency at short horizons. The practical lesson is that simpler, more constrained strategies are preferable when N/T is large.

8 References

Table 1: Median out-of-sample monthly Sharpe ratios by strategy and sample size.

Strategy	T=60	T=120	T=240	T=600
Equal weight (1/N)	0.102	0.102	0.102	0.102
Min. variance	0.021	0.037	0.043	0.046
Volatility parity	0.102	0.102	0.102	0.102
Factor portfolio	0.102	0.102	0.102	0.102
Plug-in tangency	0.006	0.010	0.019	0.032